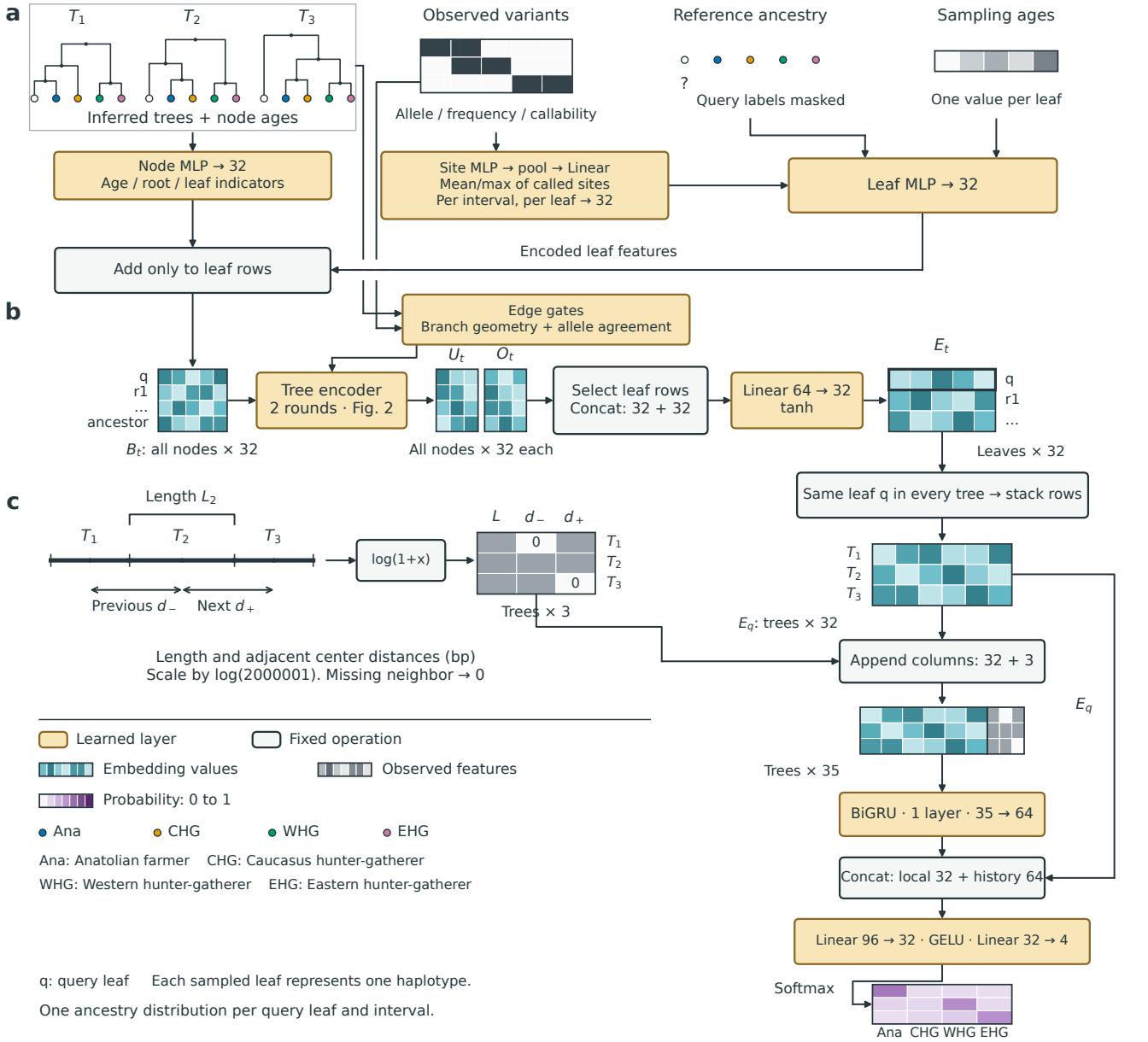
a
T_1
T_2
T_3
Inferred trees + node ages
Observed variants
Allele / frequency / callability
Reference ancestry
?
Query labels masked
Sampling ages
One value per leaf
Node MLP → 32
Age / root / leaf indicators
Site MLP → pool → Linear
Mean/max of called sites
Per interval, per leaf → 32
Leaf MLP → 32
Encoded leaf features
Add only to leaf rows
b
Edge gates
Branch geometry + allele agreement
U_t
O_t
E_t
q
r1
...
ancestor
Tree encoder
2 rounds · Fig. 2
Select leaf rows
Concat: 32 + 32
Linear 64 → 32
tanh
q
r1
...
B_t: all nodes × 32
All nodes × 32 each
Leaves × 32
c
Length L_2
T_1
T_2
T_3
Previous d_-
Next d_+
log(1+x)
L
d_-
d_+
0
0
T_1
T_2
T_3
Trees × 3
Length and adjacent center distances (bp)
Scale by log(2000001). Missing neighbor → 0
Same leaf q in every tree → stack rows
T_1
T_2
T_3
E_q: trees × 32
Append columns: 32 + 3
E_q
Trees × 35
BiGRU · 1 layer · 35 → 64
Concat: local 32 + history 64
Linear 96 → 32 · GELU · Linear 32 → 4
Softmax
Ana
CHG
WHG
EHG
Learned layer
Fixed operation
Embedding values
Observed features
Probability: 0 to 1
Ana
CHG
WHG
EHG
Ana: Anatolian farmer
CHG: Caucasus hunter-gatherer
WHG: Western hunter-gatherer
EHG: Eastern hunter-gatherer
q: query leaf
Each sampled leaf represents one haplotype.
One ancestry distribution per query leaf and interval.